

# Naive Bayes Foundations: Conditional Probability

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Leads to:** Bayes' theorem (Video 85, coming).

## 1. Overview

**Key point:** The probability of A given B,  $P(A | B)$ , is the probability of A once we know B has happened. It equals  $P(A \cap B) / P(B)$ .

The next Notes build the **Naive Bayes** classifier, one of the simplest and fastest classification algorithms, widely used for text such as spam filtering. It rests on a few ideas from probability: conditional probability, independent events, and Bayes' theorem. This Note covers the first.

**Conditional probability** answers questions of the form “how likely is A, now that we know B is true?”. It is used throughout probability and machine learning, and Bayes' theorem is built directly on it.

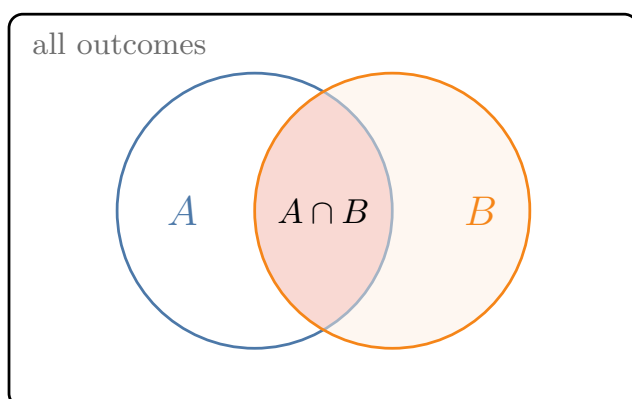
## 2. The definition

**Key point:**  $P(A | B) = P(A \cap B) / P(B)$ , as long as  $P(B)$  is not zero.

For two events A and B:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

- $P(A | B)$  is read “the probability of A given B”.
- $A \cap B$  (A intersection B) means “both A and B happen”.



$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

once B is known to have happened,  
B becomes the whole world:  
what share of B is also A?

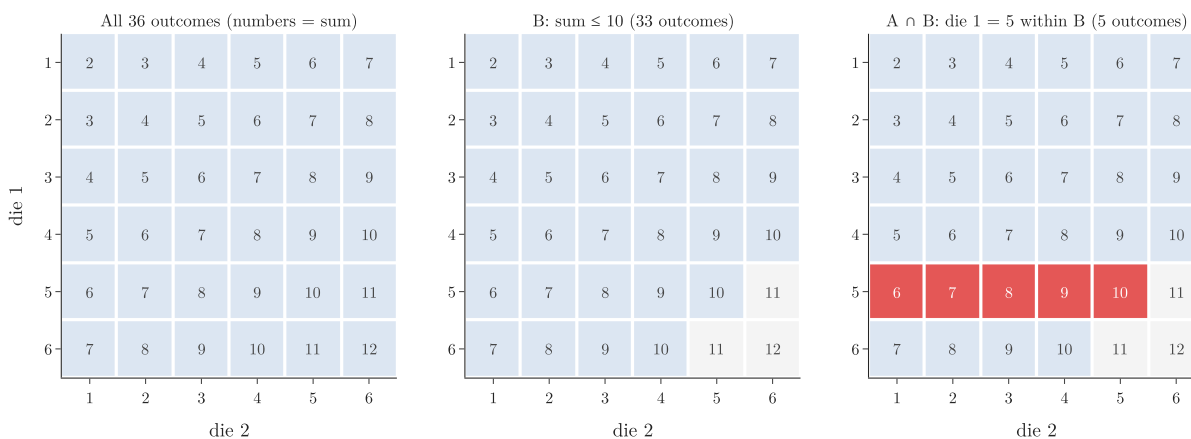
**Figure 1:** Conditional probability as a share of B

The idea (Figure 1): once we know  $B$  has happened, every outcome outside  $B$  is ruled out.  $B$  becomes the whole world, and the question is what share of it also belongs to  $A$ .

### 3. An example with two dice

**Key point:** Rolling two dice has 36 equally likely outcomes. Conditioning on “sum  $\leq 10$ ” shrinks the space to 33 outcomes.

Roll two dice together. Each outcome is a pair, (die 1, die 2):  $(1, 1), (1, 2), \dots, (6, 6)$ . There are  $6 \times 6 = 36$  outcomes, all equally likely. This set of all outcomes is the **sample space**.



**Figure 2:** The 36 outcomes, the event  $B$ , and  $A \cap B$

Figure 2 lays the sample space out as a grid, with the sum of the two dice in each cell.

#### 3.1 Two ordinary probabilities

**Key point:**  $P(\text{die 1} = 5) = 6/36$ ;  $P(\text{sum} \leq 10) = 33/36$ .

With equally likely outcomes, a probability is the number of favourable outcomes divided by the total.

- **A: die 1 shows 5.** That is the whole fifth row: 6 outcomes.  $P(A) = 6/36 = 1/6$ .
- **B: the sum is at most 10.** Only three outcomes break it:  $(5, 6)$ ,  $(6, 5)$  and  $(6, 6)$ , with sums 11, 11 and 12. So  $P(B) = 33/36 = 11/12$  (Figure 2, middle).

#### 3.2 A conditional probability by counting

**Key point:** Count only inside  $B$ : 5 of its 33 outcomes have die 1 = 5, so  $P(A \mid B) = 5/33$ .

Now ask: **what is the probability that die 1 shows 5, given that the sum is at most 10?**

Since B is known to have happened, only its 33 outcomes are possible. This is the **reduced sample space**. Inside it, die 1 shows 5 in the outcomes (5, 1), (5, 2), (5, 3), (5, 4) and (5, 5); the sixth, (5, 6), has sum 11 and is ruled out (Figure 2, right). So

$$P(A \mid B) = \frac{5}{33} \approx 0.152$$

Knowing B has lowered the chance of A slightly, from  $6/36 \approx 0.167$ , because B ruled out one of A's outcomes.

### 3.3 The same answer from the formula

**Key point:**  $P(A \cap B) = 5/36$  and  $P(B) = 33/36$ ; dividing gives  $5/33$ .

Using the formula instead, with the full sample space of 36:

- $P(A \cap B)$ : die 1 is 5 **and** the sum is at most 10. These are the same 5 outcomes, so  $P(A \cap B) = 5/36$ .
- $P(B) = 33/36$ .

$$P(A \mid B) = \frac{5/36}{33/36} = \frac{5}{33}$$

The 36s cancel, leaving exactly the count in the reduced sample space. The formula and the counting say the same thing.

**Python:** Counting all 36 outcomes.

```
from fractions import Fraction
from itertools import product
```

```
outcomes = list(product(range(1, 7), repeat=2))
```

```
B = [o for o in outcomes if sum(o) <= 10] # 33 outcomes
```

```
A_and_B = [o for o in B if o[0] == 5] # 5 outcomes
```

```
Fraction(len(A_and_B), len(B)) # 5/33
```

```
A simulation of a million rolls gives 0.1512, close to  $5/33 = 0.1515$ .
```

## 4. $P(A \mid B)$ is not $P(B \mid A)$

**Key point:** Swapping the roles changes the answer:  $P(\text{sum} \leq 10 \mid \text{die 1} = 5) = 5/6$ , not  $5/33$ .

The order matters. Given that die 1 shows 5, the reduced space is its 6 outcomes, of which 5 have a sum of at most 10:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{5/36}{6/36} = \frac{5}{6}$$

So  $P(B \mid A) = 5/6$  while  $P(A \mid B) = 5/33$ . Confusing the two is a classic mistake. Bayes' theorem (two Notes later) is exactly the rule that converts one into the other.

## 5. Summary

- Conditional probability:  $P(A | B) = P(A \cap B)/P(B)$ .
- Knowing B shrinks the sample space to B; count A's share inside it.
- Two dice:  $P(\text{die 1} = 5 | \text{sum} \leq 10) = 5/33$ , against  $P(\text{sum} \leq 10 | \text{die 1} = 5) = 5/6$ .

## 6. Key terms

| Term                        | Meaning                                                                       |
|-----------------------------|-------------------------------------------------------------------------------|
| Sample space                | The set of all possible outcomes of an experiment                             |
| Event                       | A set of outcomes, such as “the sum is at most 10”                            |
| Intersection ( $A \cap B$ ) | The event that both A and B happen                                            |
| Conditional probability     | The probability of an event given that another event has happened: $P(A   B)$ |
| Reduced sample space        | The outcomes that remain possible once the condition is known                 |
| Naive Bayes                 | A classification algorithm based on Bayes' theorem (later Notes)              |